Intelligent Remote Bed Elevation Control System

Smooth Servo Kinematics with Patient-Condition Adaptive Control

Institution: Padmabhooshan Vasantraodada Patil Institute of Technology (PVPIT), Budhgaon, Sangli

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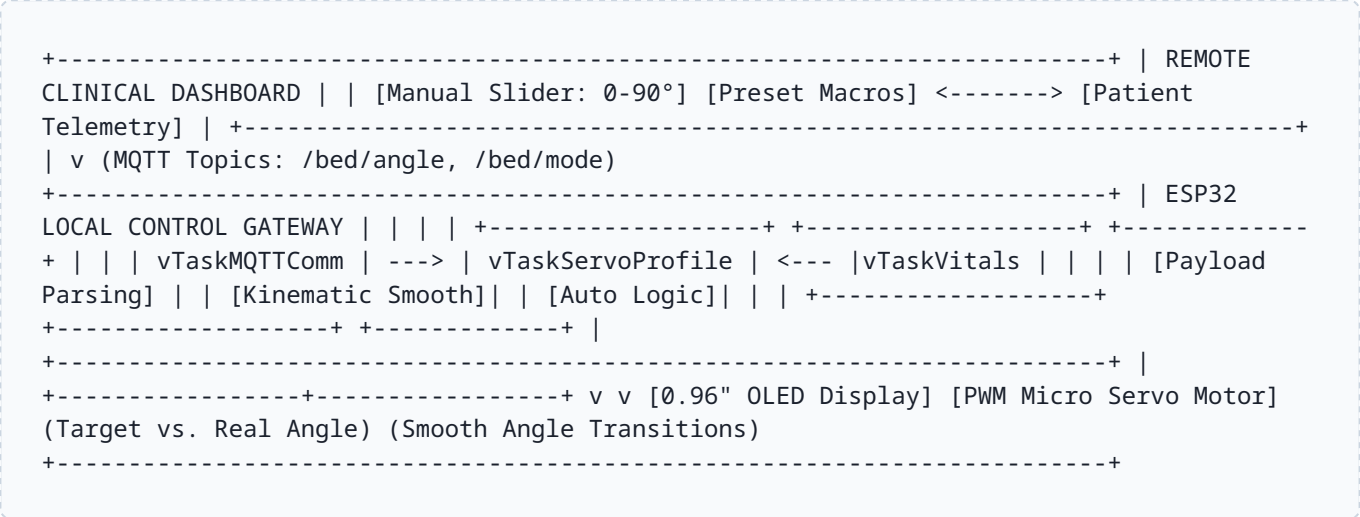
1. System Architecture Diagram

The system layout orchestrates a closed-loop positioning architecture where position updates are initiated via a cloud dashboard (manual slider or pre-set buttons) or derived locally via automated clinical feedback. A positional profiling thread intercepts instructions, converting abrupt adjustments into smooth acceleration ramps before updating the physical servo actuator.



2. FreeRTOS Task Structure Explanation

To implement smooth motion profiles concurrently without disrupting critical network handshakes or data parsing loops, mechanical calculations and device drivers are isolated into independent FreeRTOS execution threads.

Task Schedule Parameters

Task Name String	Priority	Periodicity / Trigger	Functional System Objective
vTaskMQTTComm	Medium (3)	200ms (Periodic)	Manages network connections, parses incoming positioning demands, and reports back status streams.
vTaskServoProfile	Highest (5)	15ms (Periodic)	Generates incremental angle step updates to ensure smooth, controlled movement.
vTaskVitalsMonitor	High (4)	500ms (Periodic)	Analyzes patient condition inputs to trigger automated structural emergency maneuvers.
vTaskDisplayUI	Low (1)	250ms (Periodic)	Drives the local OLED animation loops, rendering active targets and safety states.
vTaskWatchdog	Highest (6)	100ms (Periodic)	Monitors thread execution flags; forces device reset into safe defaults if locks occur.

Synchronization Primitives

- **Kinematic Position Mutex:** Encapsulates target and current angle data structures, preventing multi-thread write conflicts when manual inputs override autonomous adjustment loops.
- **Emergency Override Semaphores:** Allows the vitals evaluation task to immediately preempt active positioning routines when critical medical events are detected.

3. System Feed Mapping Table

Communication utilizes distinct topics to organize user control operations, preset macro selections, and real-time device tracking indicators.

MQTT Topic Endpoint Namespace	Data Format	Direction	Description / Structural Context
`/bed/angle/set`	Integer (0 - 90)	Subscribe	Manual target inclination command requested via the dashboard slider interface.
`/bed/angle/current`	Integer (0 - 90)	Publish	Real-time mechanical angle coordinates currently achieved by the actuator array.
`/bed/mode/select`	String ID	Subscribe	Triggers preset clinical macro positions: `"SLEEP"`, `"BREATHING"`, or `"EMERGENCY"`.
`/bed/system/status`	JSON Packet	Publish	Telemetry logs containing execution metrics, motion profiles, and active safety states.

4. Remote UI Control Dashboard Design

The system control panel combines linear input widgets with quick-action macro triggers and active state visualizers.

[Hospital Ward Bed Positioning Control Console View]

• Manual Inclination Controller: Linear slider widget mapped strictly from 0° (Flat) to 90° (Vertical Elevation).

• Quick-Action Macro Panel: Three macro execution buttons for clinical positioning presets (Sleeping, Breathing Support, Emergency).

• Kinematic Tracking Graph: Live dual-line chart contrasting User Target Angle against Real-Time System Angle.

5. Clinical Position Preset Modes

The control layout establishes predefined mechanical presets configured to meet common patient care requirements:

- **Sleeping Mode (10° elevation):** Provides a slight incline to prevent acid reflux and minimize lower back tension while maintaining optimal sleeping comfort.
- **Breathing Support Mode (45° elevation):** Semi-Fowler's position configuration designed to improve chest expansion, maximize oxygen intake, and ease respiratory distress.
- **Emergency Mode (90° elevation):** Brings the patient into a full upright position instantly during critical clinical procedures or to assist recovery teams during choking or acute aspiration hazards.

6. Kinematic Smoothing & Step-Limiting Logic

To avoid sudden, jerky mechanical movements that could cause patient discomfort or distress, the system applies a time-sampled step-limiting algorithm inside the high-frequency 15ms execution block.

Instead of driving the servo directly to a new target angle, the controller limits changes to a maximum angular velocity ($\omega_{max} = 3^\circ ext{ per second}$). The target path is broken down into small, incremental sub-steps:

$$ heta_{real}(n) = heta_{real}(n-1) + ext{sign}\left(heta_{target} - heta_{real}(n-1) ight) \cdot \min\left(\left| heta_{target} - heta_{real}(n-1)\right|, \; \omega_{max} \cdot \Delta t ight)$$

This approach smooths sudden large changes (e.g., jumping from 0° to 90°) into a safe, linear ramp, ensuring smooth and comfortable transitions.

```
// Kinematic Position Profiling Driver Implementation
uint8_t process_smooth_trajectory(uint8_t target, uint8_t current, float dt) {
    float max_step = 3.0 * dt; // Maximum angular change parameter rule (3 degrees/sec)
    int16_t positional_error = (int16_t)target - (int16_t)current;

    if (positional_error > 0) {
        current += (max_step > positional_error) ? positional_error : max_step;
    } else if (positional_error < 0) {
        current -= (max_step > -positional_error) ? -positional_error : max_step;
    }
    return current;
}
```

7. Patient-Condition Adaptive Control

The system features an autonomous feedback routine that allows it to automatically respond to changes in patient health parameters:

- **Telemetry Evaluation:** The `vTaskVitalsMonitor` thread continuously processes incoming data streams from connected clinical hardware (such as oxygen saturation or respiration rate monitors).
- **Autonomous Adjustments:** If patient oxygen levels dip below 92% (indicating respiratory distress), the system bypasses manual control and automatically switches to **Breathing Support Mode (45°)** to assist lung expansion.
- **Override Hierarchy:** Autonomous safety overrides are given absolute priority over manual dashboard adjustments. Manual inputs are temporarily ignored until vital signs stabilize or medical staff explicitly acknowledge and clear the alert flag.

8. Safety Constraints & System Failsafes

To operate reliably within busy medical environments, the system incorporates multiple structural and software protections:

- **Mechanical Angular Limits:** Hardcoded boundaries within the edge driver strictly enforce the 0° to 90° range, completely preventing out-of-bounds servo signals that could damage hardware or strain the bed frame.
- **System Lockout Detection:** The dedicated `vTaskWatchdog` runs on an independent internal countdown timer. If a firmware glitch or connection failure disrupts the smooth motion generator task, the watchdog automatically cuts power to the servo driver, locking the bed securely in its current position to prevent uncontrolled movement.